



Cambridge International AS & A Level

PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

May/June 2020

MARK SCHEME

Maximum Mark: 30

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam. The answer column of the mark scheme shows the proposed basis on which Examiners would award marks for this exam. Where appropriate, this column also provides the most likely acceptable alternative responses expected from students. Examiners usually review the mark scheme after they have seen student responses and update the mark scheme if appropriate. In the June series, Examiners were unable to consider the acceptability of alternative responses, as there were no student responses to consider.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i> • Incorrect responses should not be awarded credit but will still count towards <i>n</i> • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form, (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations

✓	Correct point Method of analysis marks in Question 1
✓ ₁₋₁₀	Additional detail marks in Question 1
X	Incorrect point
^	Omission
BOD	Benefit of the doubt
NBOD	No benefit of the doubt given
ECF	Error carried forward
P	Defining the problem marks in Question 1 Power of ten error in Question 2
M0	Methods of data collection marks in Question 1

Question	Answer	Marks
1	Defining the problem	
	A is the independent variable and k is the dependent variable or vary A and measure k	1
	keep N <u>constant</u>	1
	Methods of data collection	
	labelled diagram of workable experiment including: • spring fixed at one end to a support • load attached to the other end of the spring • labelled load	1
	method to measure mass or weight of load: use top-pan balance to measure mass or newton meter to measure weight	1
	use of a micrometer/calipers to determine t and rule/calipers to measure the diameter of the spring	1
	method to measure extension, e.g. labelled ruler drawn parallel to spring, equilibrium position and displaced position indicated and x indicated or description of use of ruler to measure equilibrium position and displaced position and difference determined	1
	Method of analysis	
	plot a graph of k against $1/A^{3/2}$ (or $A^{-3/2}$) or equivalent e.g. $\lg k$ against $\lg A$	1
	relationship valid if a straight line passing through the origin is produced (for $\lg k$ against $\lg A$, relationship valid if a straight line with gradient $-3/2$)	1
	$\beta = \frac{\text{gradient} \times N}{\rho t^4}$ [for $\lg k$ against $\lg A$, $\beta = 10^{\text{y-intercept}} \times N / (\rho t^4)$]	1

Question	Answer	Marks
1	Additional detail including safety considerations	6
D1	use safety goggles/safety screen <u>to prevent injury to eyes from (moving) spring/load</u> or use cushion/sand box <u>in case load falls</u>	
D2	keep t constant	
D3	$k = \frac{mg}{x}$ or $\frac{F}{x}$	
D4	use of set square when taking measurements to determine extension of spring	
D5	repeat measurement of t <u>along wire/spring</u> and average	
D6	repeat measurement of diameter D of spring (to determine A) <u>in different directions</u> and average	
D7	use of $A = \frac{\pi D^2}{4}$	
D8	method to ensure clamped rule to measure extension is vertical, e.g. correctly positioned set square indicated at right angles between the rule and the horizontal surface or plumb line shown in appropriate position	
D9	method to determine the density of the wire or additional detail on construction of coil	
D10	method to determine the mean diameter of the spring, e.g. subtract t from external diameter of spring	

Question	Answer	Marks							
2(a)	$\text{gradient} = \frac{-1}{CR}$ $y\text{-intercept} = \ln \frac{Q_0}{C}$	1							
2(b)	<table><tr><td>$\ln (V / V)$</td></tr><tr><td>1.82 or 1.825</td></tr><tr><td>1.53 or 1.526</td></tr><tr><td>1.22 or 1.224</td></tr><tr><td>0.96 or 0.956</td></tr><tr><td>0.69 or 0.693</td></tr><tr><td>0.34 or 0.336</td></tr></table>	$\ln (V / V)$	1.82 or 1.825	1.53 or 1.526	1.22 or 1.224	0.96 or 0.956	0.69 or 0.693	0.34 or 0.336	1
$\ln (V / V)$									
1.82 or 1.825									
1.53 or 1.526									
1.22 or 1.224									
0.96 or 0.956									
0.69 or 0.693									
0.34 or 0.336									
	Absolute uncertainties in $\ln V$ from ± 0.03 or ± 0.04 to ± 0.13 or ± 0.14 or ± 0.15 .	1							
2(c)(i)	Six points plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square.	1							
	Error bars in $\ln V$ plotted correctly. All error bars to be plotted. Length of bar must be accurate to less than half a small square and symmetrical.	1							
2(c)(ii)	Line of best fit drawn. Points must be balanced. Do not accept top point to bottom point. Line must pass between (4.0, 1.6) and (5.0, 1.6) and between (26.5, 0.5) and (28.0, 0.5)	1							
	Worst acceptable line drawn (steepest or shallowest possible line that passes through all the error bars). All error bars must be plotted.	1							

Question	Answer	Marks
2(c)(iii)	Gradient determined with clear substitution of data points into $\Delta y / \Delta x$. Distance between data points must be at least half the length of the drawn line. Gradient must be negative.	1
	uncertainty = (gradient of line of best fit – gradient of worst acceptable line) or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept read from y-axis to less than half a small square or y-intercept determined from substitution into $y = mx + c$.	1
2(d)(i)	C determined using gradient and C and Q_0 given to two or three significant figures. Correct substitution of numbers required. $C = \frac{-1}{39 \times 10^3 \times \text{gradient}} = \frac{-1}{39 \times 10^3 \times (\text{c})(\text{iii})}$	1
	Q_0 determined using y-intercept. $Q_0 = C \times e^{y\text{-intercept}} = C \times e^{(\text{c})(\text{iv})}$	1
	C determined using gradient and Q_0 determined using y-intercept and dimensionally correct units for C (F or $\text{s}\Omega^{-1}$) and Q_0 (C or $\text{V s}\Omega^{-1}$ or As).	1
2(d)(ii)	Absolute uncertainty in C. $\Delta C = \left(0.05 + \frac{\Delta \text{gradient}}{\text{gradient}} \right) \times C$	1

Question	Answer	Marks
2(e)	V determined from (d)(i) (or (c)(iii) and (c)(iv)) with correct substitution shown and correct power of ten. $V = \frac{Q_0}{C} \times e^{\frac{-60}{CR}} = e^{y\text{-intercept}} \times e^{(\text{gradient} \times 60)}$ or $\ln V = -(t / RC) + \ln (Q_0 / C) = -(60 / 39\,000) \times \mathbf{(d)(i)} + \ln (Q_0 / C)$ $\ln V = 60 \times \text{gradient} + y\text{-intercept}$ $\ln V = 60 \times \mathbf{(c)(iii)} + \mathbf{(c)(iv)}$	1